

Lecture 3

The Anatomy of LM386

Prof Peter YK Cheung
Imperial College London

URL: www.ee.ic.ac.uk/pcheung/teaching/EE2_CAS/
E-mail: p.cheung@imperial.ac.uk

What you should know already?

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EE1 Analogue Electronics

C-E Amplifier Revisited Quiescent Analysis

- KVL on input side

$$V_{BIAS} = I_B R_B + V_{BE}$$

and $V_{BE} \approx 0.7 \text{ V}$, $I_E =$

$$\Rightarrow I_E \approx \frac{(V_{BIAS} - V_{BE})}{R_E}$$

- As before:

$$V_E = I_E R_E$$

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BJT Current Mirror

INPUT SIDE



- For transistors identical apart from area

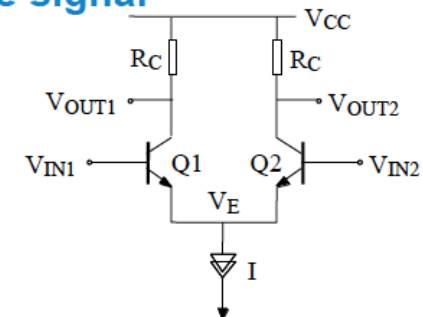
$$I \approx I_{REF}$$

where we have used $I_S \propto A$, and

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The Differential Pair – large signal

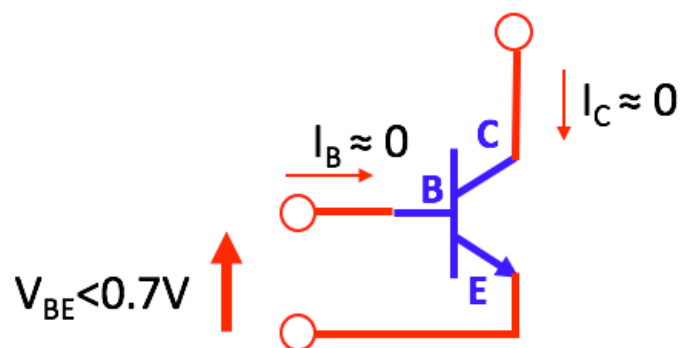


- If transistors are matched, then:

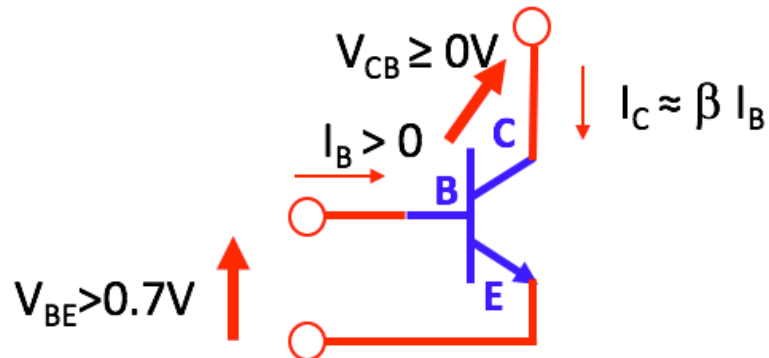
$$I_{C1} = \frac{I}{1 + \exp(-V_D/V_T)} \quad I_{C2} = \frac{I}{1 + \exp(V_D/V_T)}$$

$V_D = (V_{IN1} - V_{IN2})$ is the DIFFERENTIAL INPUT VOLTAGE

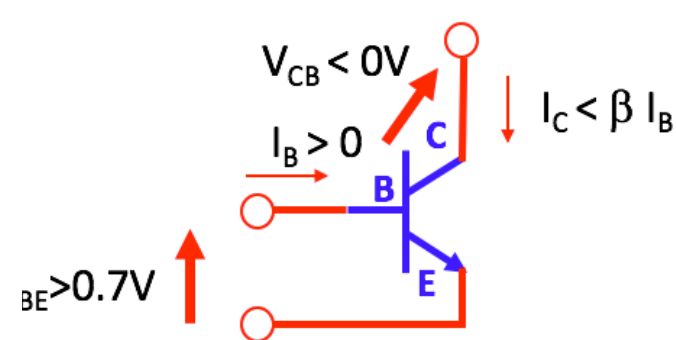
Rules of thumb – Bipolar Junction Transistor (BJT)



- ❖ $V_{BE} < 0.7V$
- ❖ Transistor OFF
- ❖ Near zero I_B and I_C (except leakage)
- ❖ Good for digital circuits

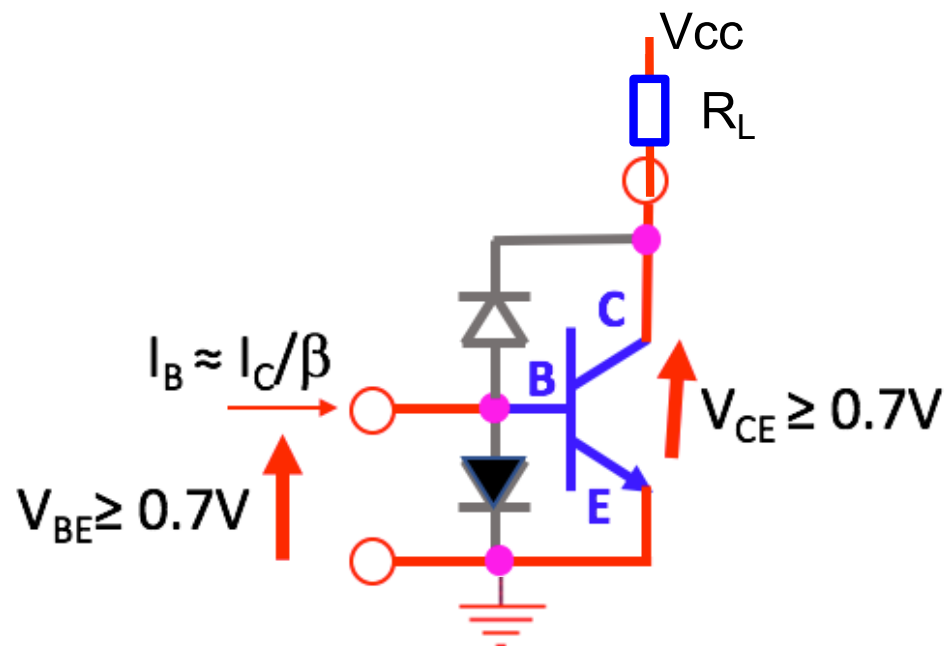


- ❖ $V_{BE} \geq 0.7V$
- ❖ BE junction diode reverse biased
- ❖ Transistor in linear/active region
- ❖ $I_C \approx \beta I_B$ applies
- ❖ Good for analogue circuits



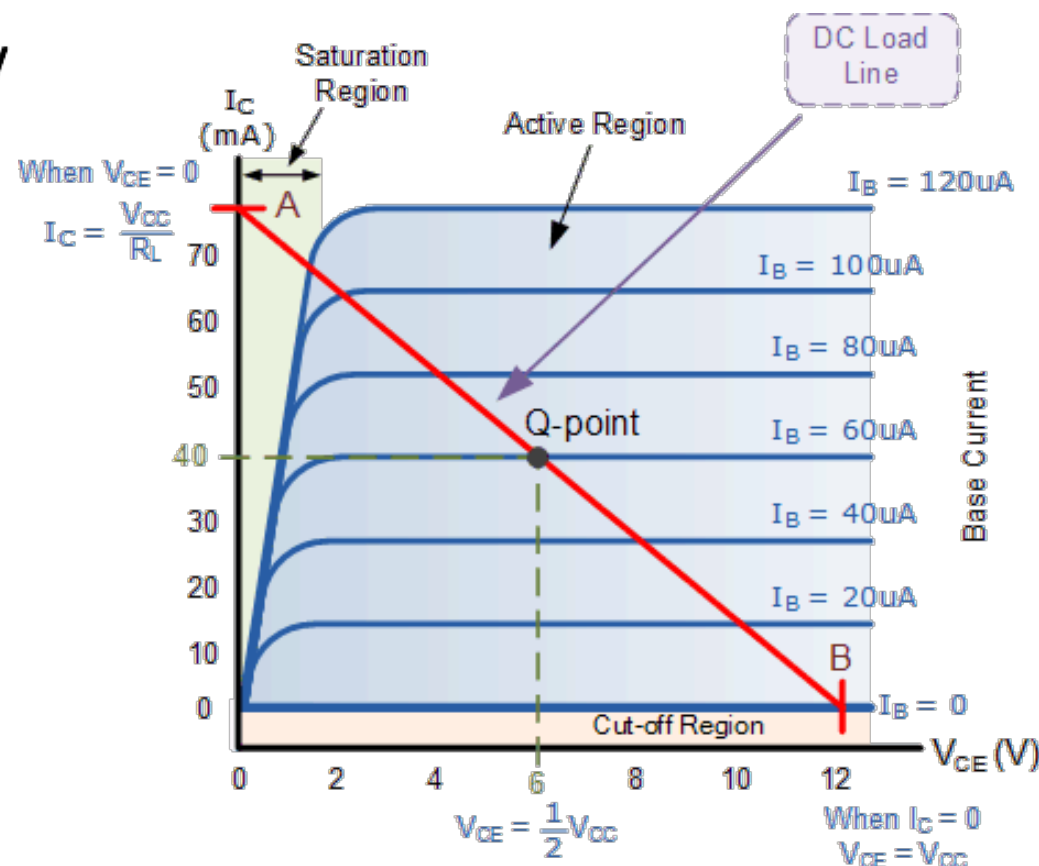
- ❖ $V_{BE} \geq 0.7V$
- ❖ BE junction diode NOT reverse biased
- ❖ Transistor in saturation region
- ❖ $I_C < \beta I_B$
- ❖ Good for digital circuits

NPN Transistor in Linear Region

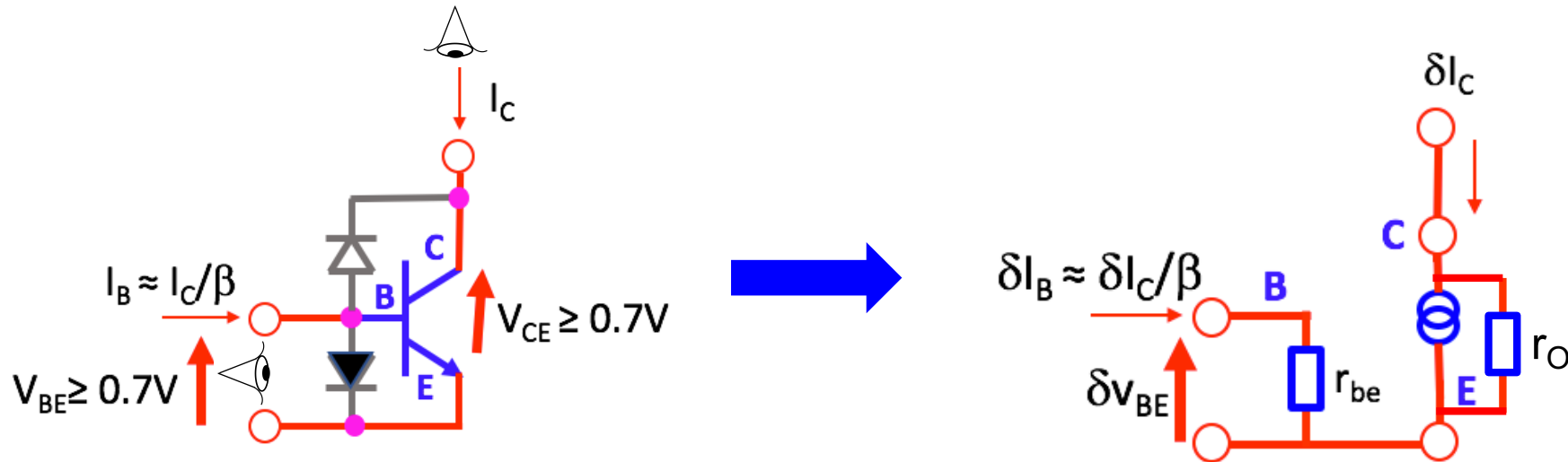


- ❖ BE junction diode forward biased (ON)
- ❖ BC junction diode OFF
- ❖ I_C is often determined by DC bias circuit
- ❖ I_B then fixed by β
- ❖ V_{CE} can take on any value between 0.3V and V_{CC}

- ❖ Quiescent (DC) condition – biasing
- ❖ Ensures transistor in linear region
- ❖ Operating condition determined by I_C
- ❖ Use large signal model



Small signal behaviour of NPN transistor



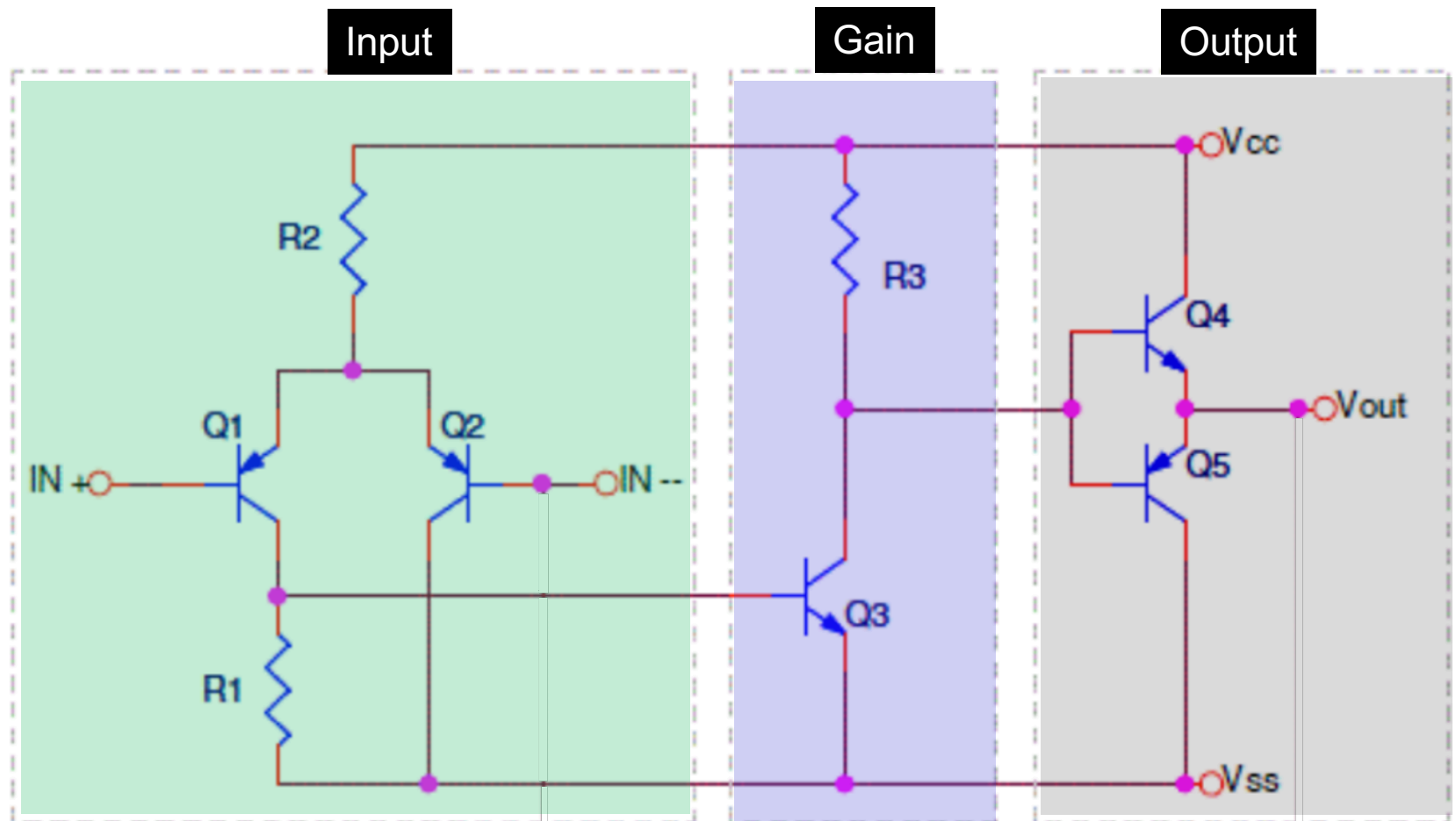
- ❖ Quiescent I_C determine operating point
- ❖ Small signal model – 2 resistors r_{be} and r_O , and a current source
- ❖ r_O large - can be omitted

- ❖ $\delta i_B = \delta v_{BE} / r_{BE}$
- ❖ $\delta i_C = g_m \delta v_{BE}$
- ❖ $\delta i_E = \delta i_B + \delta i_C$

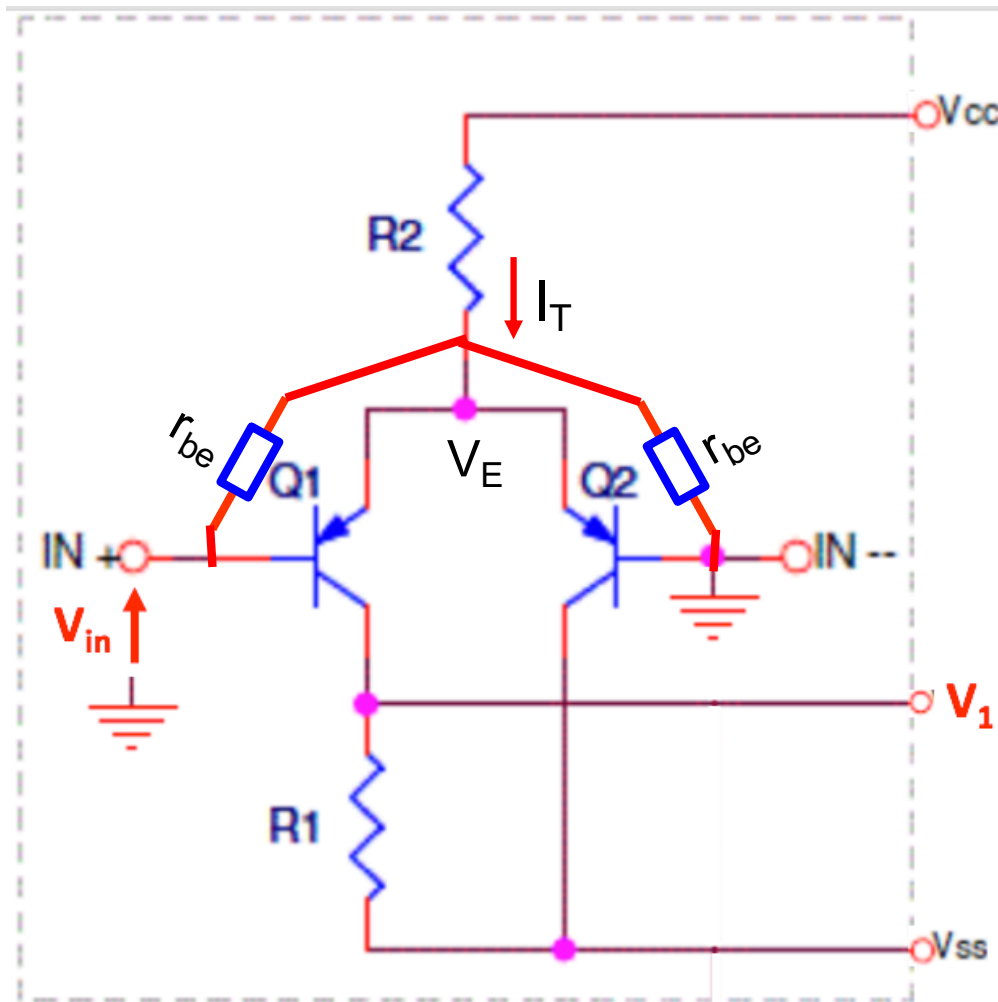
- ❖ current gain: $\beta = \delta i_C / \delta i_B$
- ❖ transconductance: $g_m = \delta i_C / \delta v_{BE} = I_C / V_T$
- ❖ Input resistance: $r_{be} = \delta v_{BE} / \delta i_B = \beta V_T / I_C$

Inside a typical op-amp

- ❖ Three stages architecture:
 1. Differential input stage – long-tail pair (Yr 1 Circuits part 2, adc_9, slides 8-14)
 2. Voltage gain stage – common emitter amp (adc_6, slides 3-7)
 3. Output drive stage – push-pull circuit



Differential Input Stage – differential gain



$$A_{V_{diff}} = \delta v_1 / \delta v_{in} = -\frac{1}{2} g_m * R_1$$

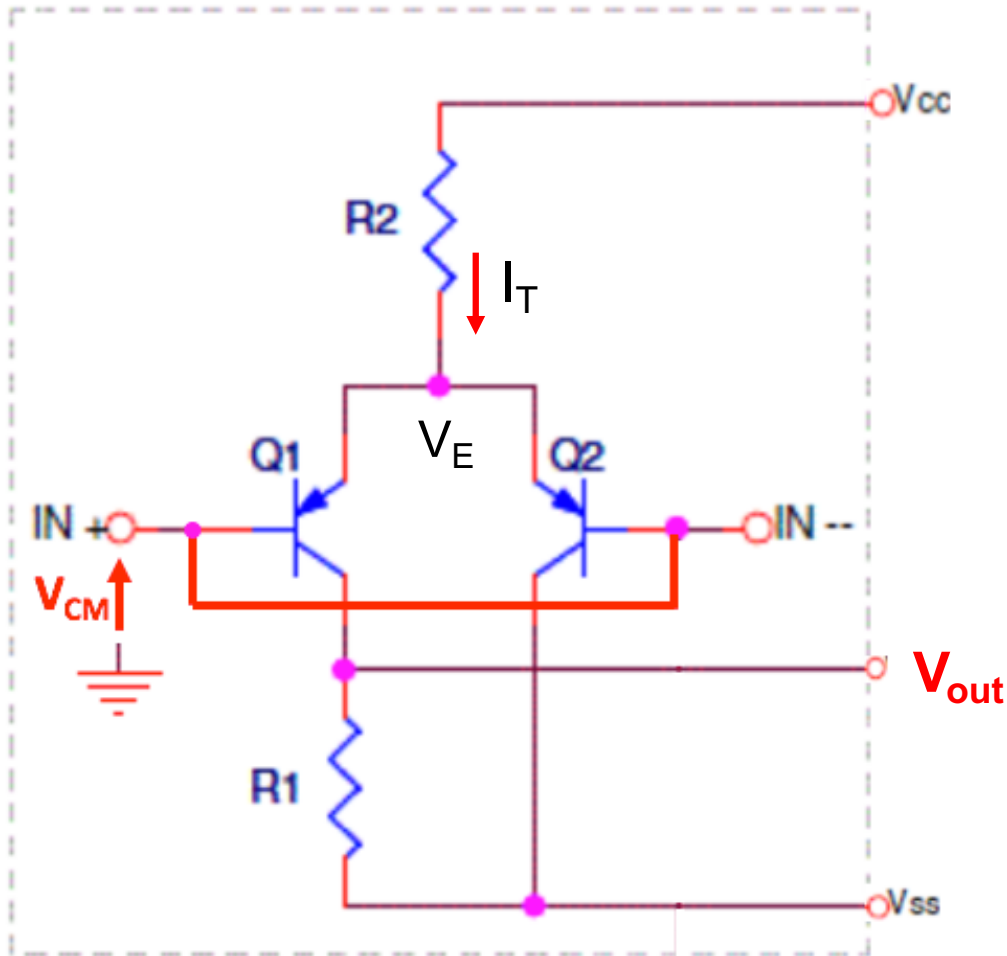
Consider $V_{in} = 0$, IN - is grounded

- ❖ $V_E \approx 0.7V$
- ❖ $I_T \approx (V_{CC} - 0.7) / R_2$
- ❖ $I_1 \approx I_2 \approx \frac{1}{2} I_T \approx \frac{1}{2} (V_{CC} - 0.7) / R_2$
- ❖ $V_1 = R_1 * I_{C1} + V_{SS}$

Apply small input signal δV_{in}

- ❖ $\delta v_E \approx \frac{1}{2} \delta v_{in}$
- ❖ $\delta i_T \approx \frac{1}{2} \delta v_{in} / R_2 \approx 0$
- ❖ $\delta i_1 \approx -\delta i_2 = -g_m * \frac{1}{2} \delta v_{in}$
- ❖ $\delta v_1 = R_1 * \delta i_1 = -R_1 * g_m * \frac{1}{2} \delta v_{in}$

Differential Input Stage – common mode gain

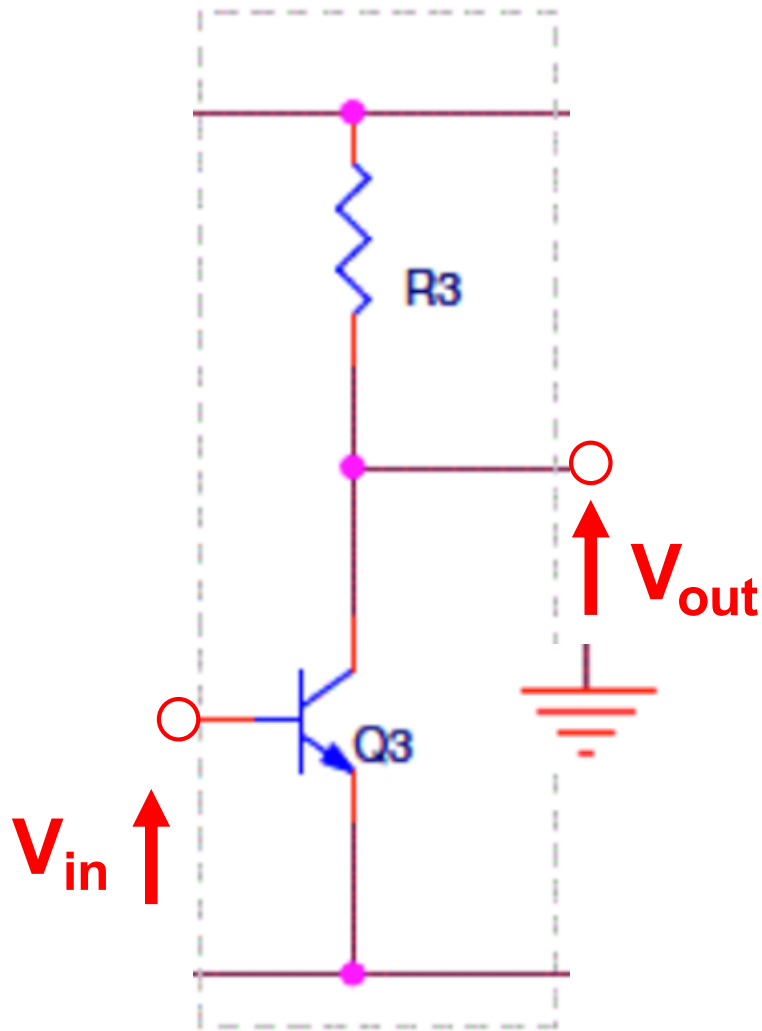


Apply v_{CM} to both inputs

- ❖ $\delta v_E \approx v_{CM}$
- ❖ $\delta i_T \approx -\delta v_E / R_2 = -v_{CM} / R_2$
- ❖ $\delta i_{C1} \approx \delta i_{C2} = \frac{1}{2} \delta i_T = -\frac{1}{2} v_{CM} / R_2$
- ❖ $\delta v_{out} \approx -R_1 / 2R_2 * v_{CM}$

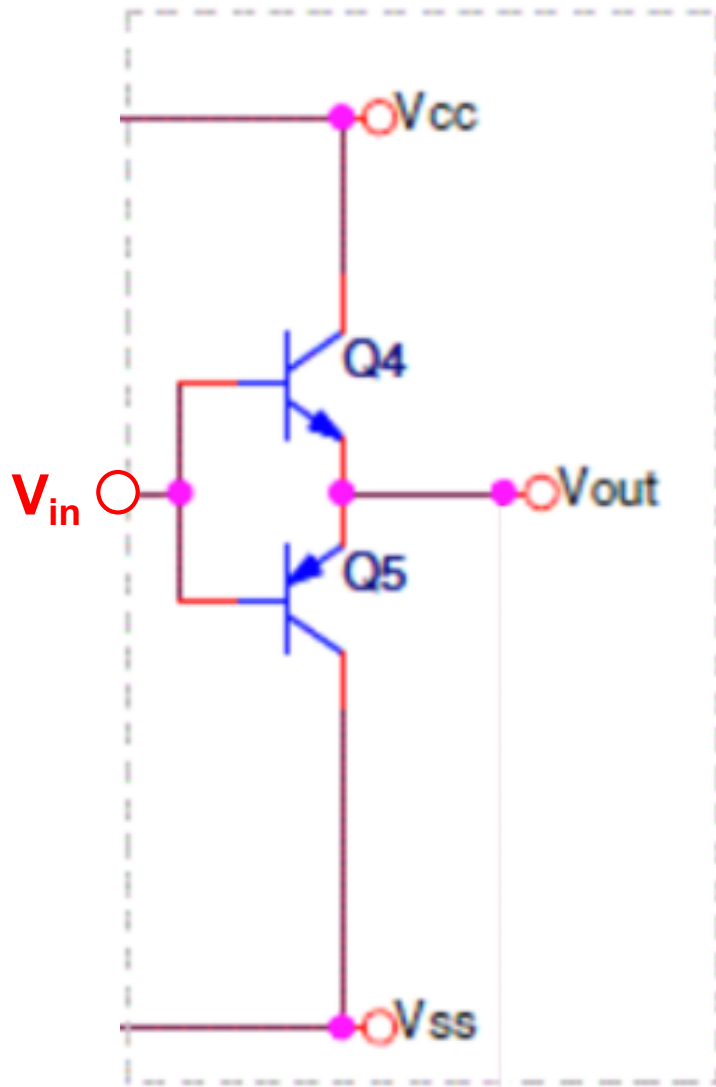
$$A_{V_{cm}} = \delta v_{out} / v_{CM} = -R_1 / 2R_2$$

Gain stage



- ❖ Common Emitter amplifier (Yr 1st ADC part 2 Lecture 6, S7)
- ❖ $\delta v_{out} = -g_m \delta v_{in} * R_3$
- ❖ $A_V = \delta v_{out} / \delta v_{in} = -g_m R_3$
- ❖ Increase g_m and R_3
- ❖ Most of the voltage gain produced by this stage

Output Stage



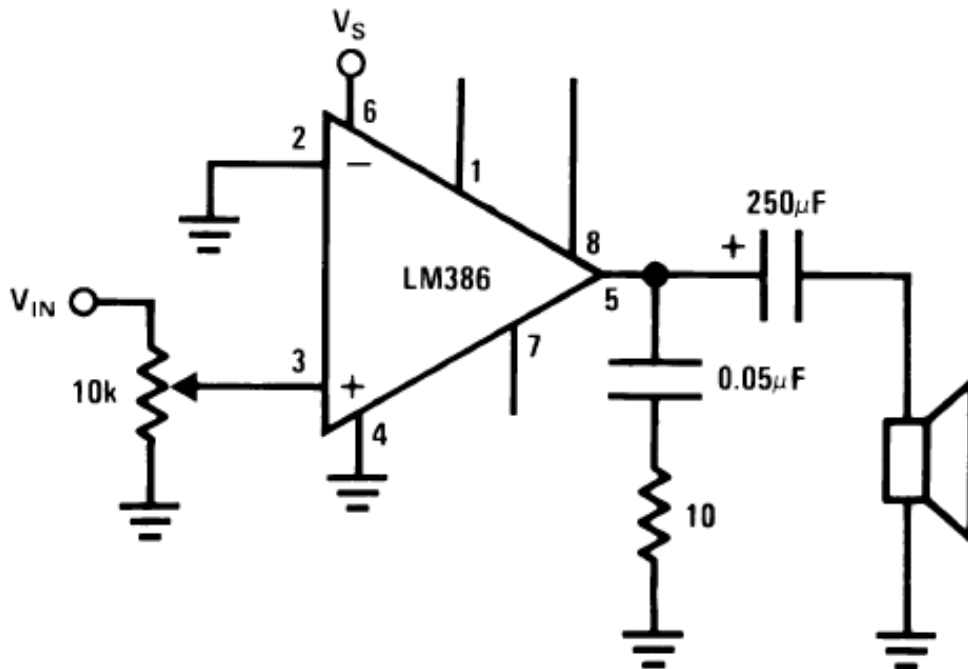
- ❖ Yr 1st ADC part 2 Lecture 6, S3-5
- ❖ Q4 is emitter follower (Common-Emitter) for sourcing current to V_{out} (PUSH)
- ❖ Q5 is another emitter follower for sinking current from V_{out} (PULL)
- ❖ This is known as a PUSH-PULL circuit
- ❖ $\delta V_{out} \approx \delta V_{in}$

Driving 8Ω speaker with LM386

LM386

Low Voltage Audio Power Amplifier

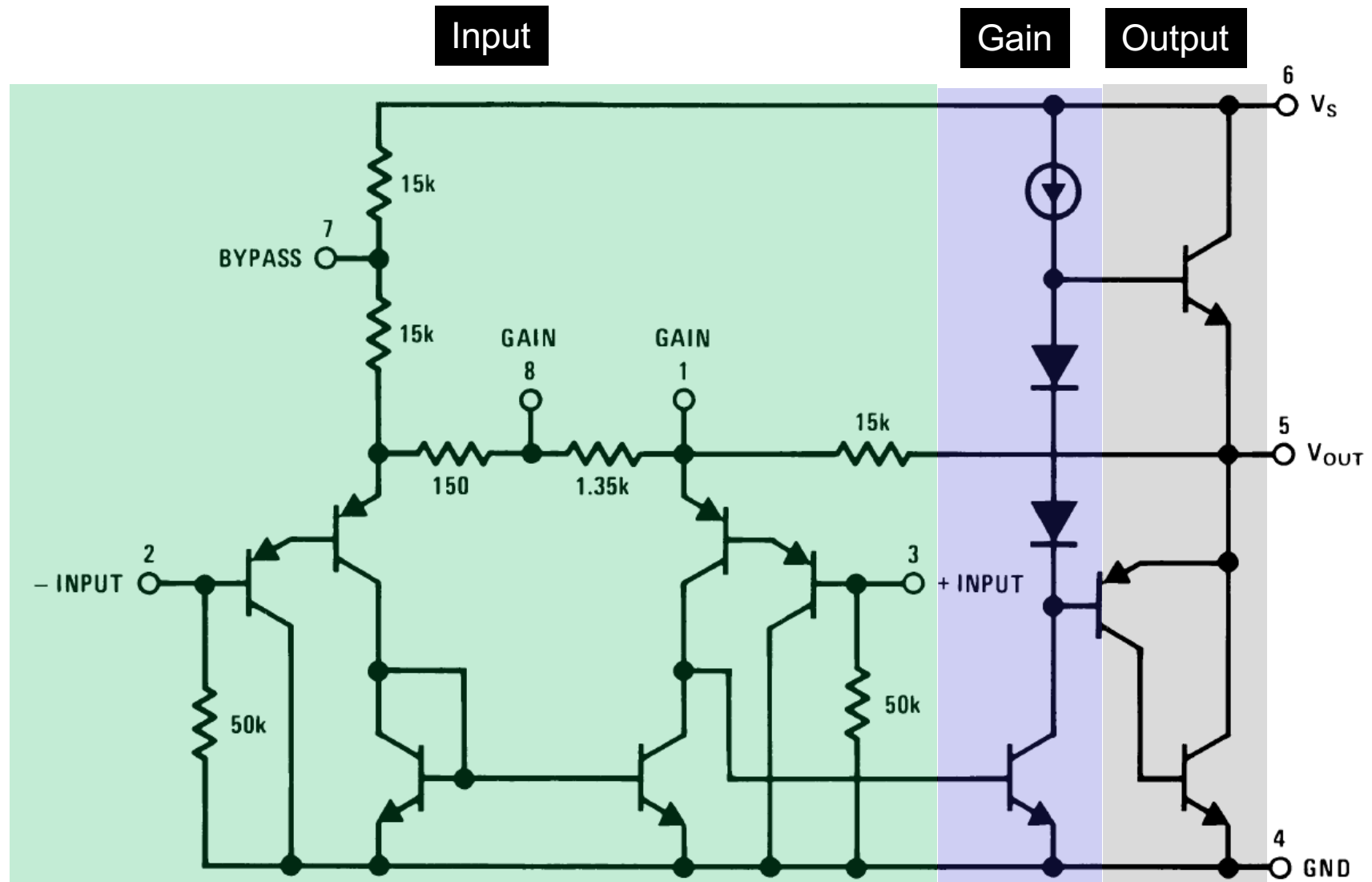
Amplifier with Gain = 20 Minimum Parts



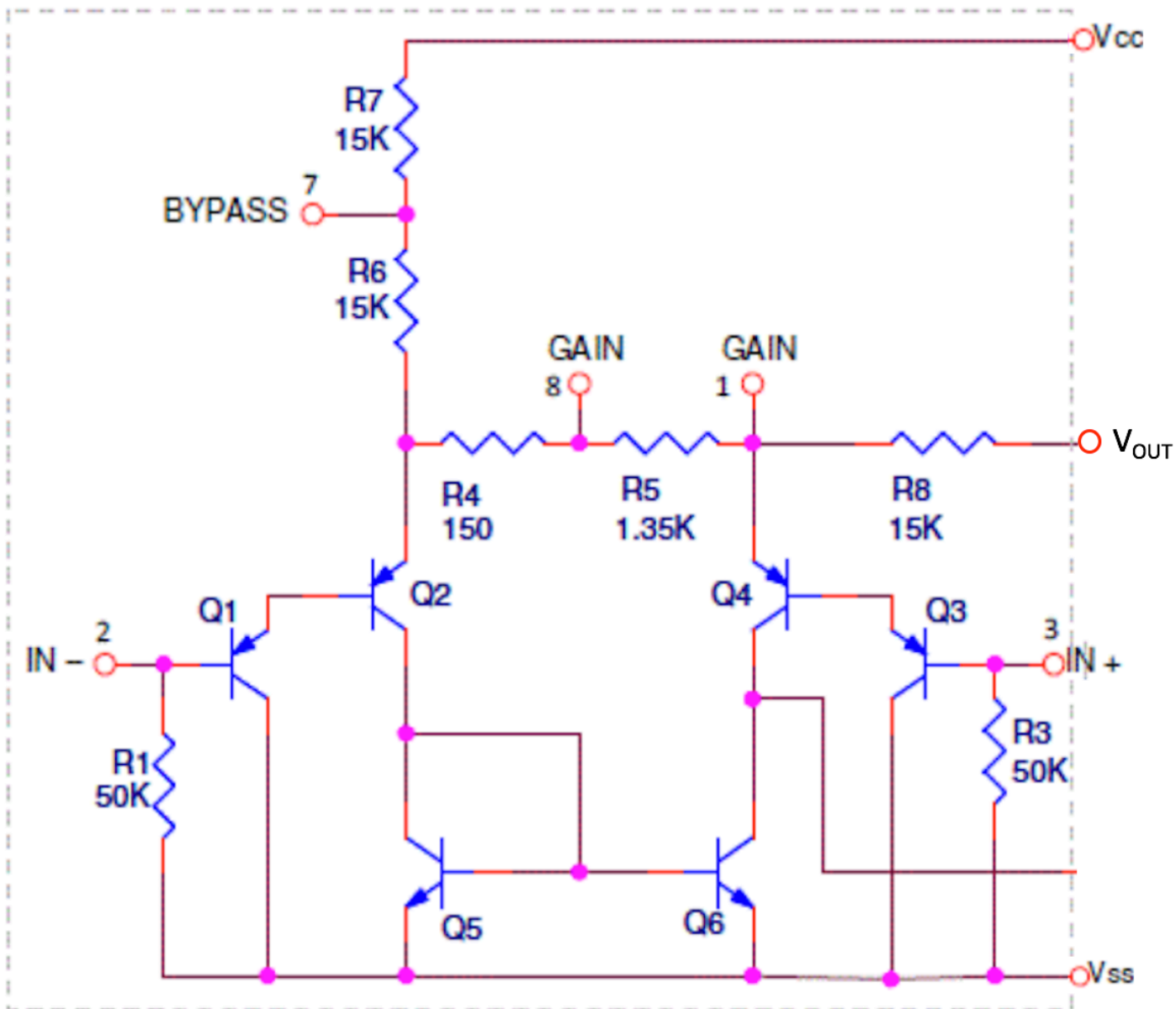
Features

- Battery operation
- Minimum external parts
- Wide supply voltage range: 4V–12V or 5V–18V
- Low quiescent current drain: 4mA
- Voltage gains from 20 to 200
- Ground referenced input
- Self-centering output quiescent voltage
- Low distortion: 0.2% ($A_V = 20$, $V_S = 6V$, $R_L = 8\Omega$, $P_O = 125mW$, $f = 1kHz$)
- Available in 8 pin MSOP package

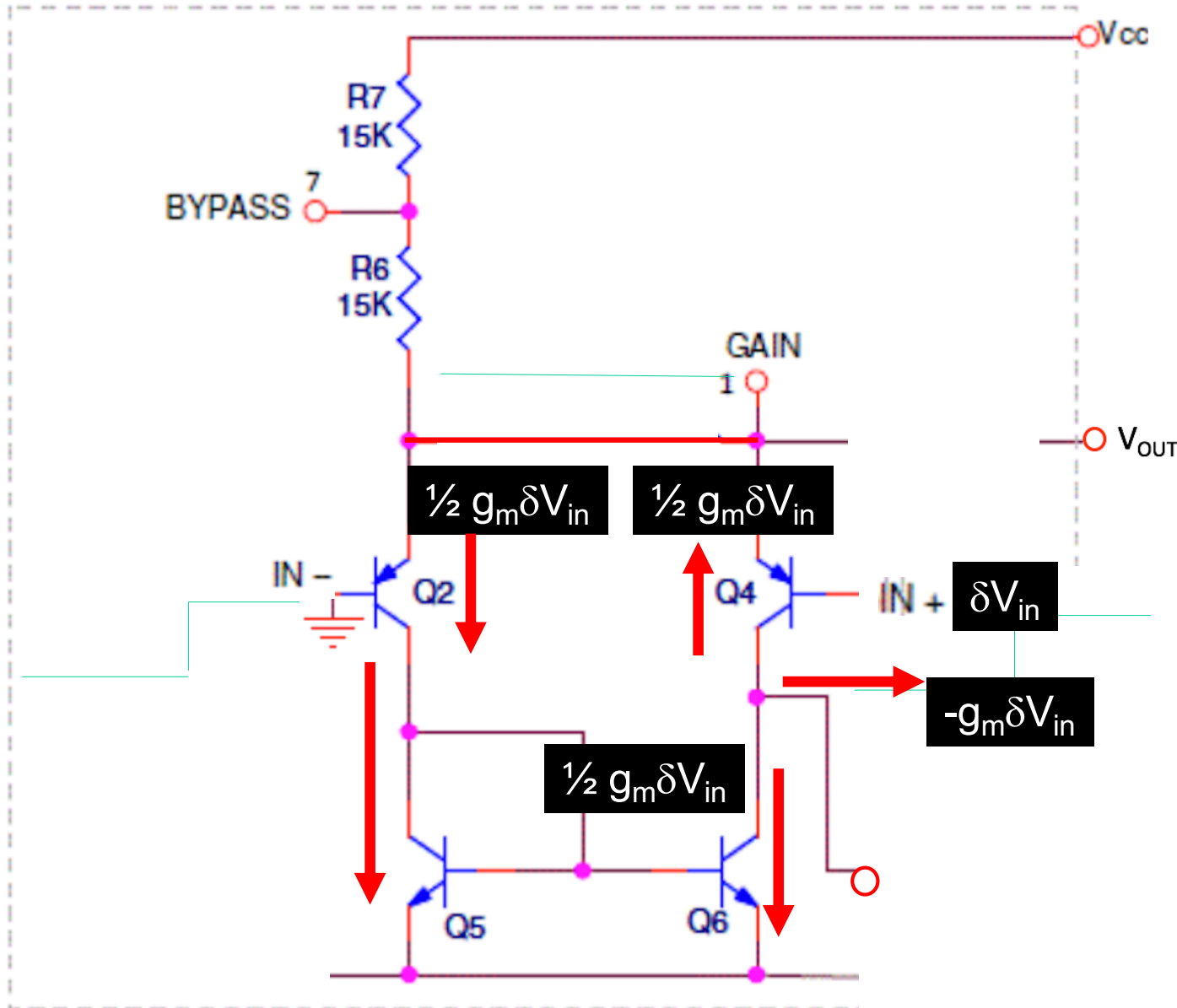
LM386 internal circuit



LM386 Input Stage (1)

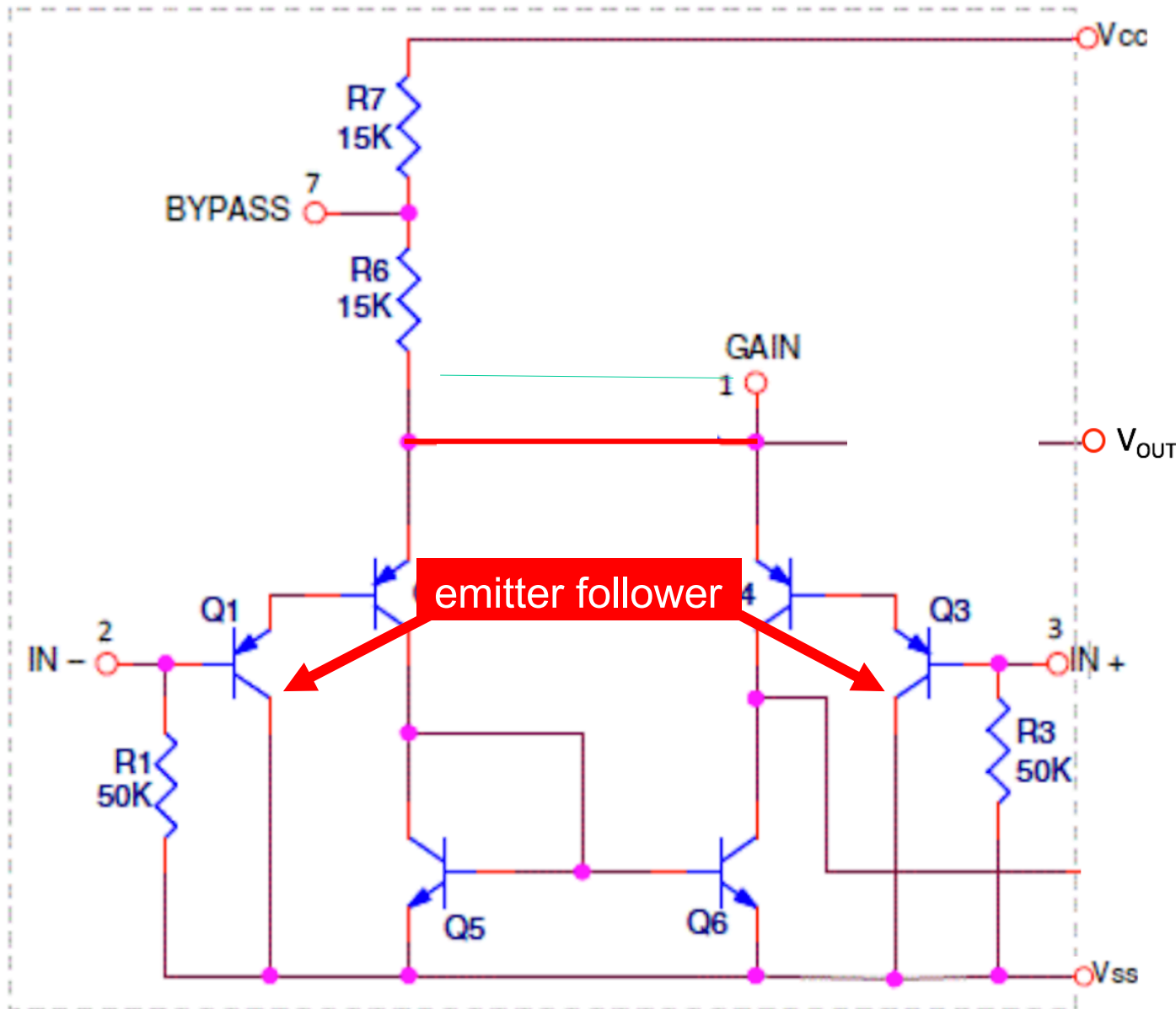


LM386 Input Stage (2)



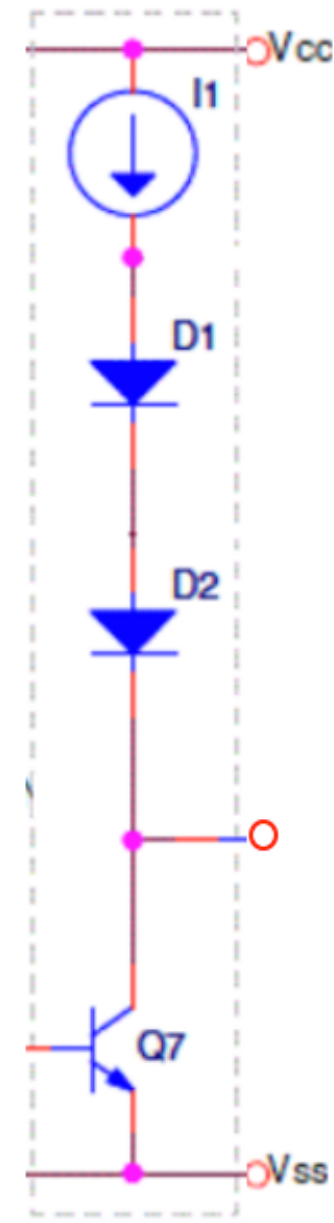
- ❖ Ignore Q1, Q3
- ❖ Ignore R4, R5
- ❖ Ignore R8
- ❖ R6, R7 – establish bias current I_T
- ❖ $\delta I_2 = -\delta I_4 = \frac{1}{2} g_m \delta V_{in}$
- ❖ Q5, Q6 current mirror: $I_2 = I_5 = I_6$
- ❖ Forces change in output current to $2 \cdot I_4$
- ❖ Hence output current changes, not voltage!

LM386 Input Stage (3)



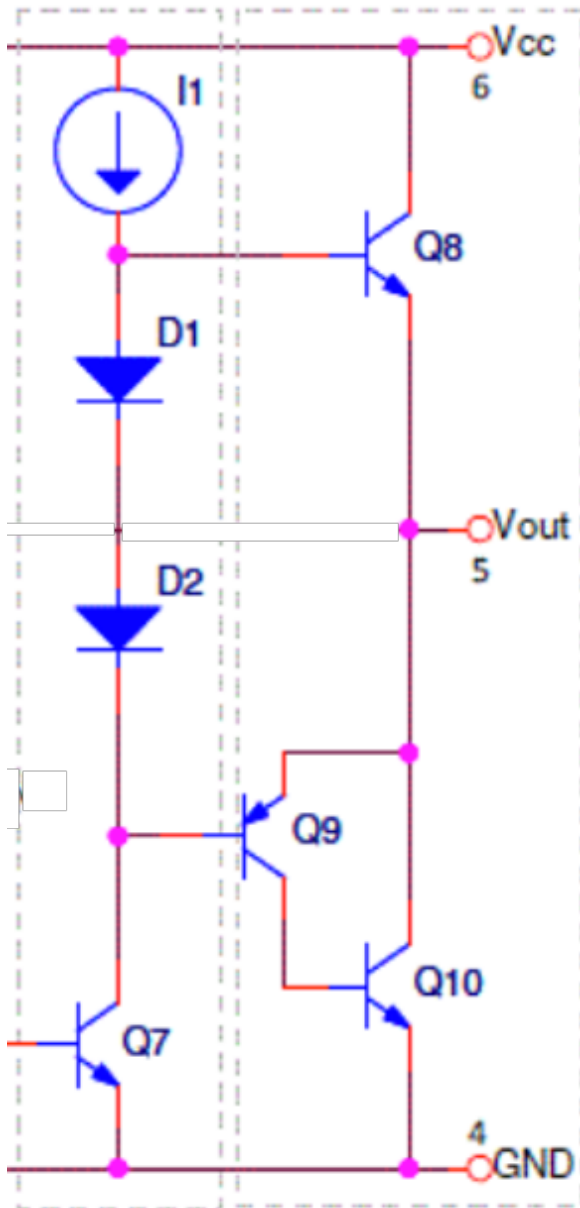
- ❖ Put back emitter followers Q1, Q3
- ❖ Set input impedance to 50k
- ❖ R1, R3 – path for base bias current
- ❖ Set DC operating point i.e. $V_E \approx 1.2 - 1.4V$
- ❖ Allows input to go slightly below GND (by around -0.4V)
- ❖ Current mirror load – high impedance = high gain

LM386 Gain Stage



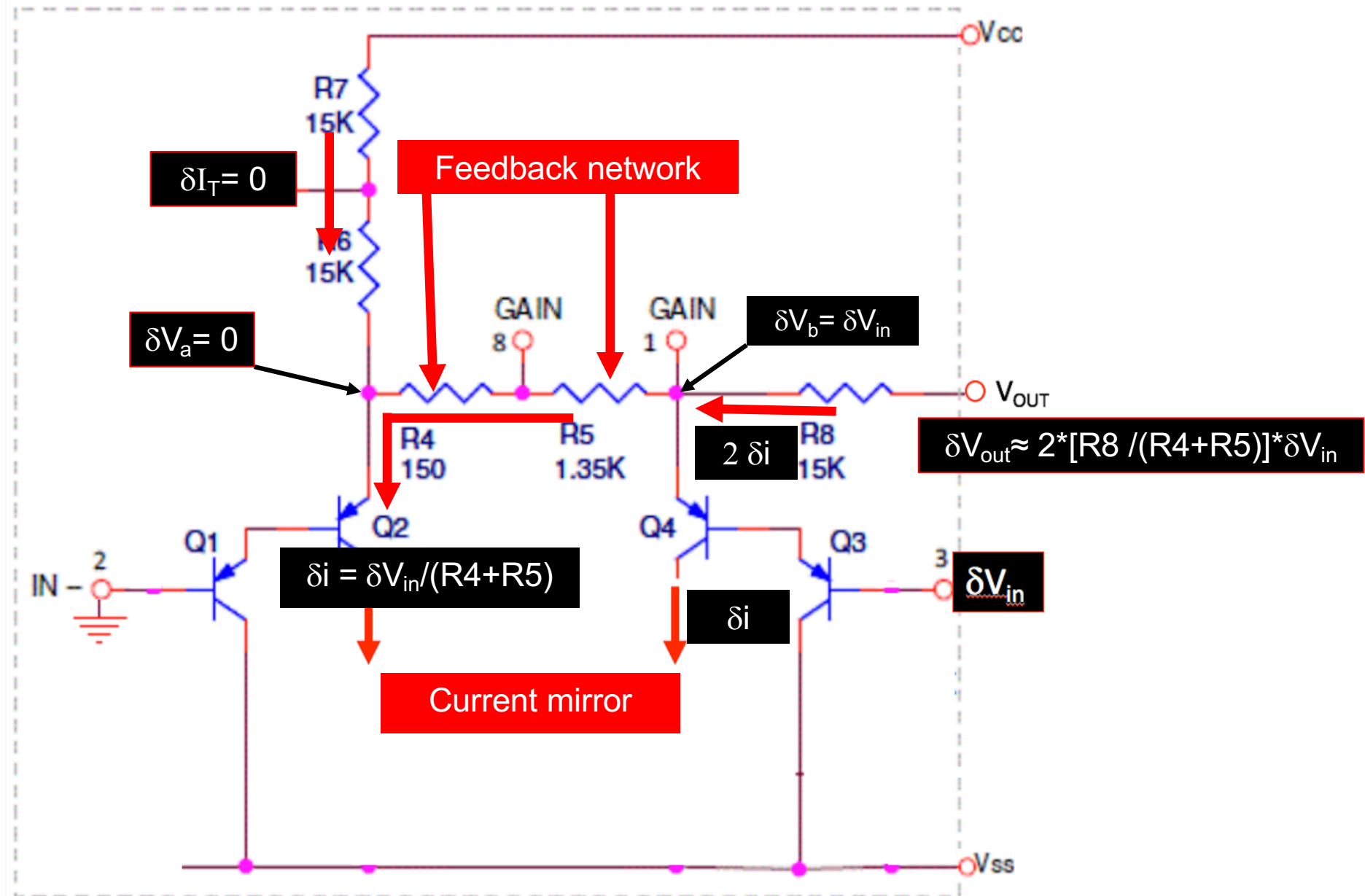
- ❖ Common emitter amplifier with active load – constant current source I_1
- ❖ Q7 g_m determined by I_1
- ❖ D1, D2 are for output stage - 1.2V constant offset
- ❖ D1, D2 has no impact on gain of 2nd stage
- ❖ Q7 collector loads a constant current source
- ❖ Gain is not through g_m , but through β
- ❖ $\delta I_{out} = \beta_7 \delta I_{B7}$

LM386 Output Stage

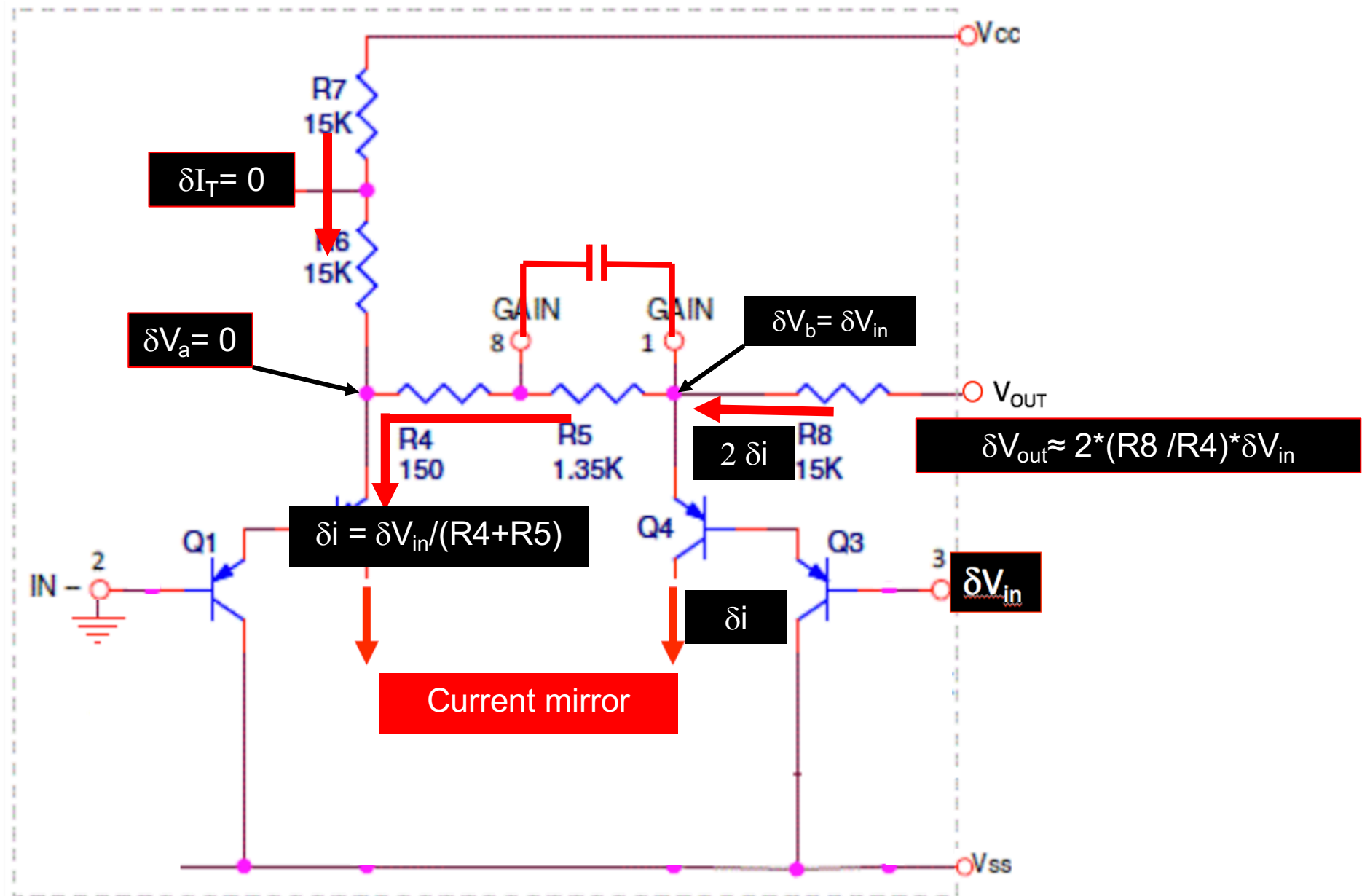


- ❖ Q_8 push emitter follower sourcing current
- ❖ Q_9 , Q_{10} pull emitter follower sinking current
- ❖ D_1 , D_2 forward bias due to I_1
- ❖ Keep Q_8 and Q_9 in linear region – reduce distortion
- ❖ Q_9 PNP has poor current gain
- ❖ Combine with Q_{10} NPN to boost current gain
- ❖ Q_9 , Q_{10} - called a “Sziklai” pair (different from Darlington pair)
- ❖ $A_v \approx 1$

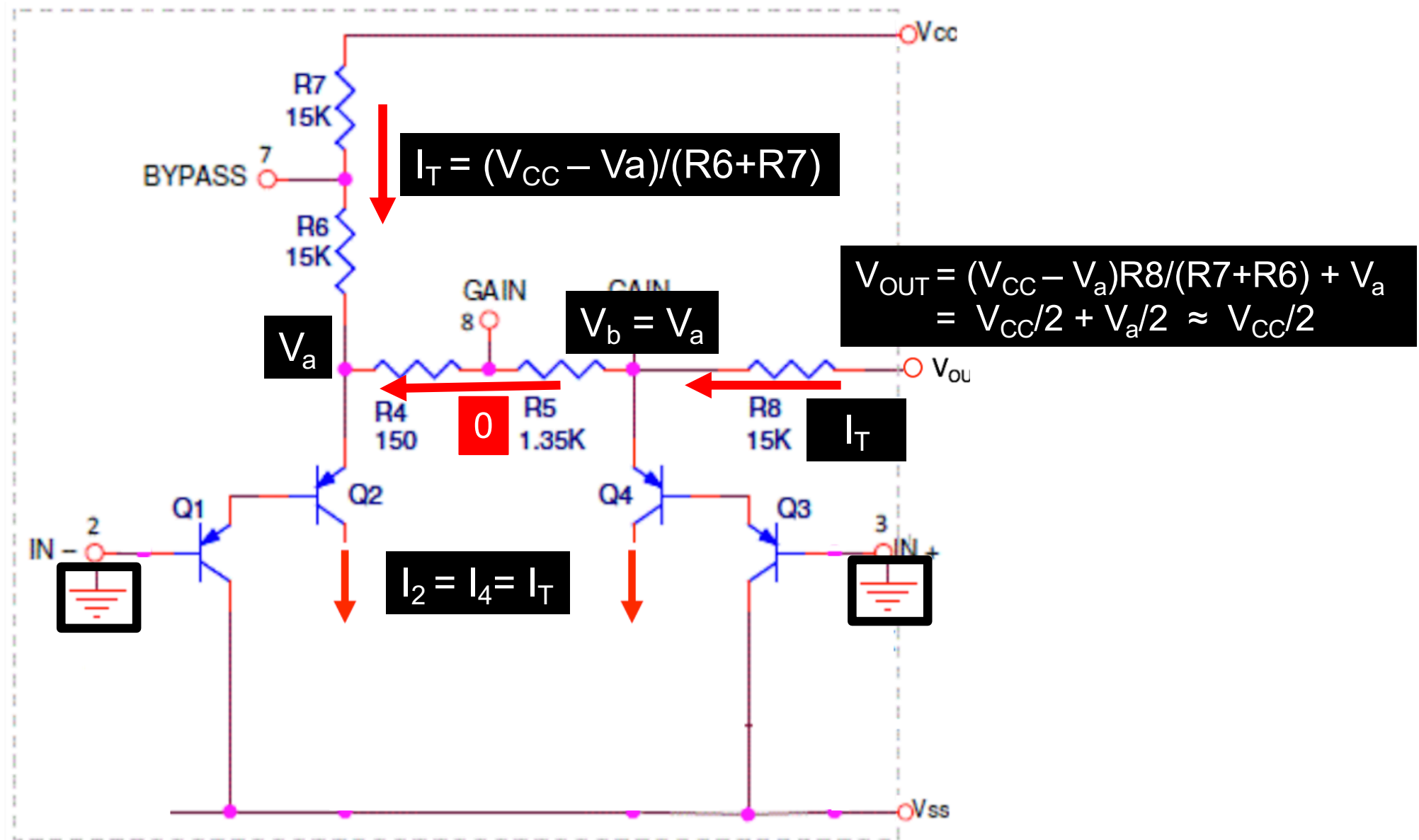
LM386 Gain is $2 \cdot R8 / (R4 + R5) \approx 20$



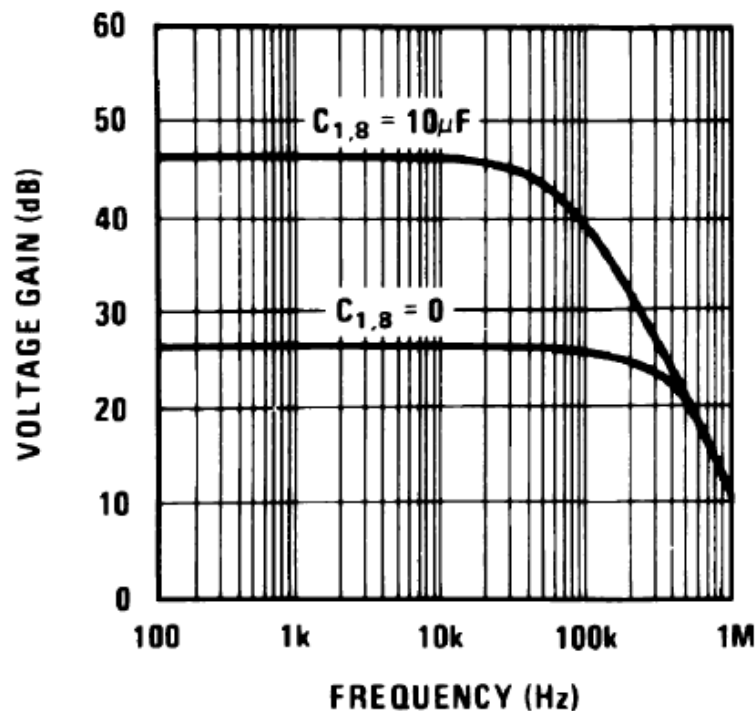
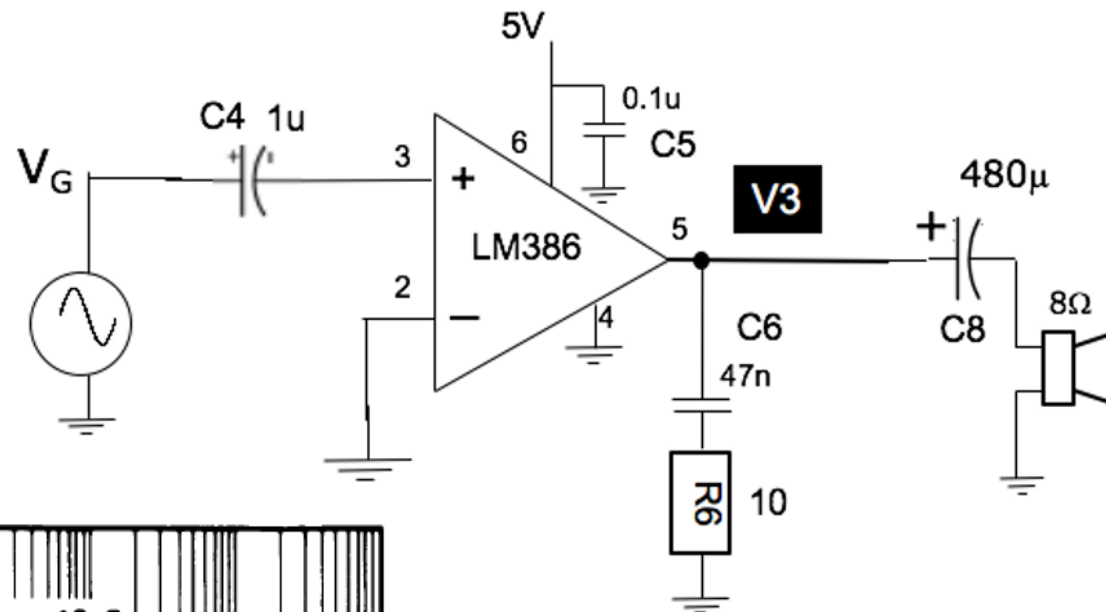
LM386 Gain is $2 \cdot R8 / R4 \approx 200$ with bypass capacitor



LM386 Vout self biasing to $V_{CC}/2$



LM386 as a practical x20 amplifier (Lab 1)



- ❖ LM386 provides x20 gain
- ❖ Drives 8 ohm speaker
- ❖ $C4$ and $C8$ – blocking DC
- ❖ $C6$, $R6$, compensate for series inductance in 8 ohm speaker